

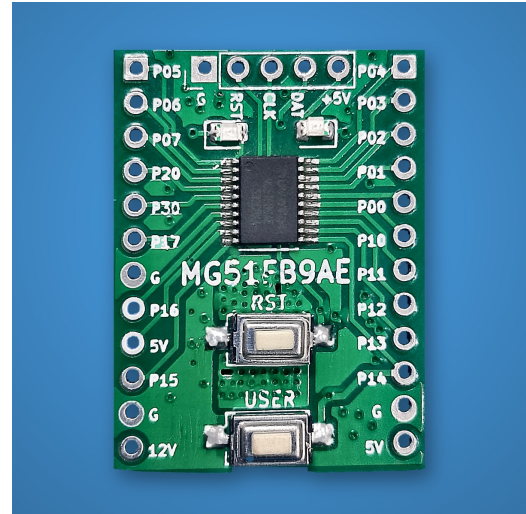


MG51 Dev Board

Compact breakout and development board for the Nuvoton MG51FB9AE 1T 8051 microcontroller

The MG51 Dev Board brings the Nuvoton NuMicro MG51FB9AE out to two 2.54 mm headers on a 24.1 x 32 mm board, so the part can be dropped straight onto a breadboard or soldered into a prototype. All 18 general purpose I/O pins are broken out, together with ground, 5 V and a 12 V input that feeds an on-board 5 V regulator. A four-pin ICP header (RST / CLK / DAT / +5 V) allows programming and debugging with a Nuvoton Nu-Link adapter, and on-board reset and user buttons plus two indicator LEDs cover the basics of bring-up without extra wiring.

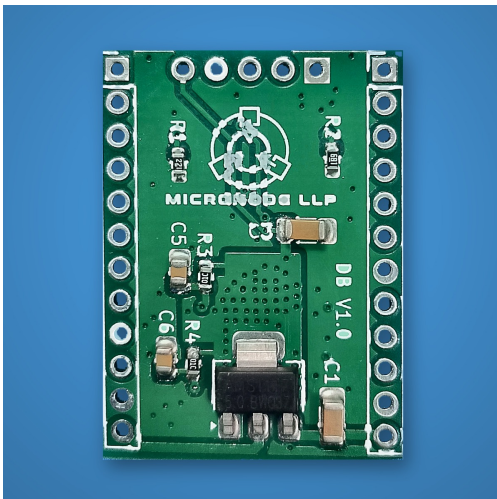
The board is aimed at engineers evaluating the MG51 for cost-sensitive designs: home appliances, LED lighting, motor control and general industrial automation. Boards ship assembled and tested, with or without the header strips soldered.



Key features

- Nuvoton MG51FB9AE, 1T 8051 core, 16 / 24 MHz
- 16 KB APROM, 1 KB XRAM, 4 KB LDROM
- 8-channel 12-bit ADC, up to 500 kSPS
- 6 PWM channels, 4 timers
- 2 x UART, 1 x SPI, 1 x I2C
- 2.4 V to 5.5 V operation, -40 to +105 C
- All 18 GPIO on 2.54 mm headers
- 5 V input, or 12 V through on-board regulator
- Reset button and user button on P1.4
- Red power LED, blue user LED on P0.5
- 4-pin ICP header for Nu-Link programming
- 24.1 x 32 mm, breadboard friendly

Bottom side



The underside carries the 5 V regulator, its input and output capacitors and the LED series resistors. No user connections are made on this side, so the board can be soldered flat onto a carrier or plugged into a socket.



1. Microcontroller

Part number	Nuvoton MG51FB9AE (NuMicro MG51 series), TSSOP-20
Core	1T 8051, instruction set compatible with MCS-51, 4 priority interrupt levels
Clock	16 MHz internal RC (default) or 24 MHz internal RC, software selectable; 10 kHz low speed RC; external clock on P3.0 (OSCIN)
Program memory	16 KB APROM, 4 KB LDROM for ISP loader, 128 B SPROM
Data memory	256 B internal RAM, 1 KB XRAM
I/O	18 GPIO, four modes per pin: quasi-bidirectional, push-pull, open-drain, input only
Analogue	8-channel 12-bit SAR ADC, up to 500 kSPS, plus internal band-gap channel
Timers	Timer 0 / 1, Timer 2 with 3-channel input capture, Timer 3 auto-reload, WDT, wake-up timer
PWM	6 channels, 16-bit, independent / complementary / synchronous modes, dead-time and brake
Serial	UART0, UART1, SPI (master or slave), I2C (100 / 400 kbps)
Supply	2.4 V to 5.5 V, -40 C to +105 C

2. Power

12 V pin	Input to the on-board linear regulator, which supplies the 5 V rail and the MCU
5 V pins	5 V rail; can be used as the supply input when no 12 V is applied
G pins	Ground, three pins available (two on the left header, one on the right)
Decoupling	Bulk and local decoupling fitted on the underside of the board

Apply either 12 V or 5 V, not both at the same time. Available current on the 5 V rail depends on the regulator dissipation at the applied input voltage.

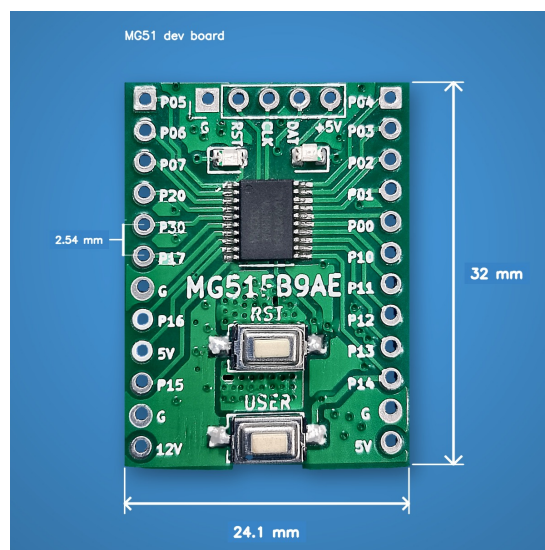
3. Controls and indicators

Designator	Type	Function
RST	Tactile switch	Pulls the MCU reset pin (P2.0 / nRESET) low while pressed
USER	Tactile switch	User input on P1.4; also broken out on the right header
Power LED	Red SMD LED	Lit whenever the 5 V rail is powered
User LED	Blue SMD LED	Driven by P0.5, also broken out on the left header

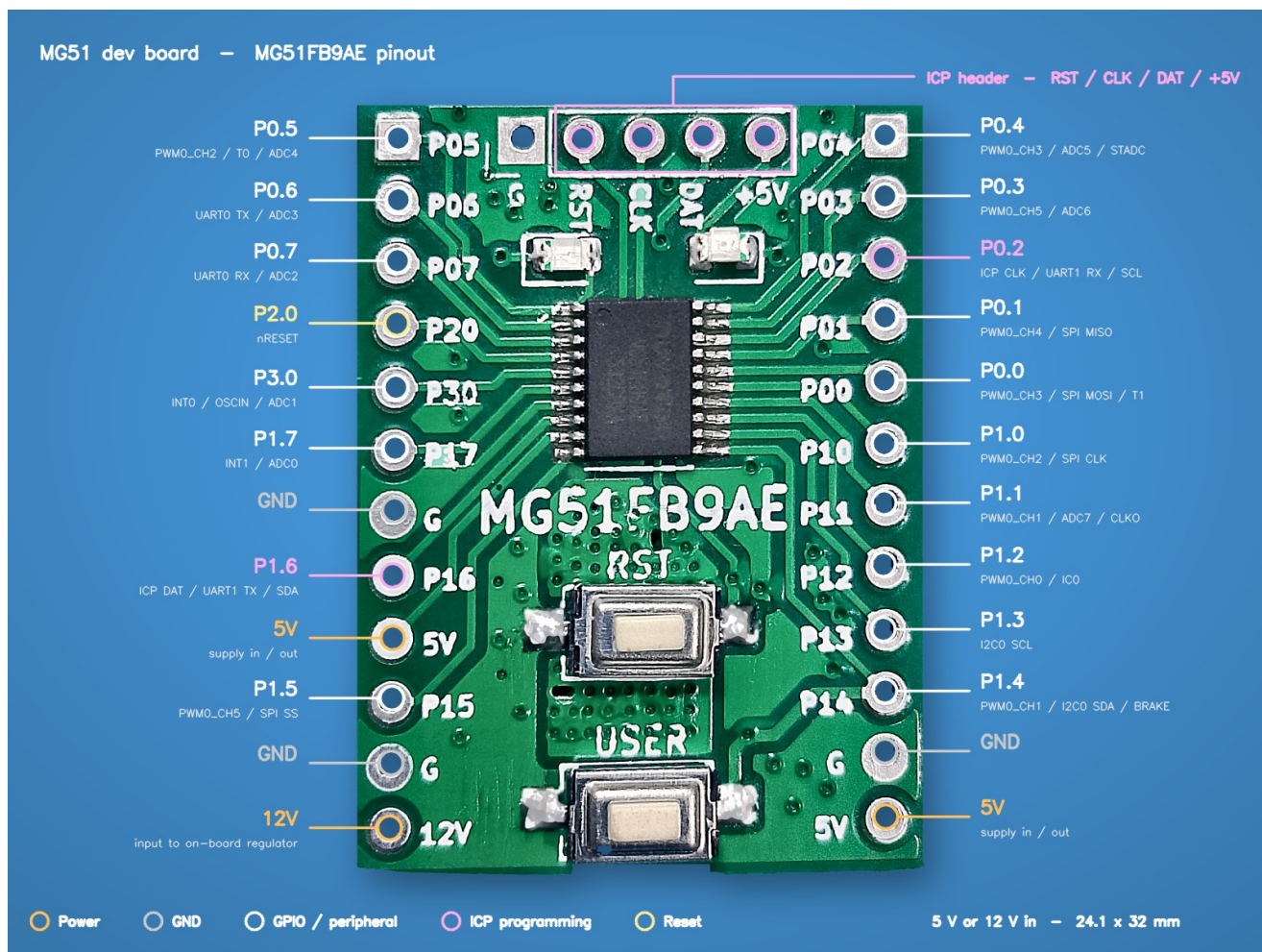


4. Mechanical

Board size	24.1 x 32 mm
Thickness	1.6 mm FR-4, green solder mask, white silkscreen
Headers	Two rows of 2.54 mm through-hole pads, 12 positions per row
Row spacing	20.32 mm between header centre lines
Mounting	No mounting holes; corner pads on the top edge



5. Pinout



Silkscreen names match the MCU port bits: P05 is P0.5, P17 is P1.7, and so on.

**Left header (top to bottom)**

Pin	Port	Alternate functions
P05	P0.5	PWM0_CH2, IC6, T0, ADC_CH4; drives the on-board blue LED
P06	P0.6	UART0_TXD, ADC_CH3
P07	P0.7	UART0_RXD, ADC_CH2
P20	P2.0	nRESET (GPIO input only when RPD is programmed)
P30	P3.0	INT0, OSCIN, ADC_CH1
P17	P1.7	INT1, ADC_CH0
G	VSS	Ground
P16	P1.6	ICE_DAT (ICP data), UART1_TXD, I2C0_SDA (remap)
5V	-	5 V rail
P15	P1.5	PWM0_CH5, IC7, SPI0_SS
G	VSS	Ground
12V	-	Regulator input

Right header (top to bottom)

Pin	Port	Alternate functions
P04	P0.4	PWM0_CH3, IC3, ADC_CH5, STADC
P03	P0.3	PWM0_CH5, IC5, ADC_CH6
P02	P0.2	ICE_CLK (ICP clock), UART1_RXD, I2C0_SCL (remap)
P01	P0.1	PWM0_CH4, IC4, SPI0_MISO
P00	P0.0	PWM0_CH3, IC3, SPI0_MOSI, T1
P10	P1.0	PWM0_CH2, IC2, SPI0_CLK
P11	P1.1	PWM0_CH1, IC1, ADC_CH7, CLK0
P12	P1.2	PWM0_CH0, IC0
P13	P1.3	I2C0_SCL, STADC (remap)
P14	P1.4	PWM0_CH1, I2C0_SDA, PWM0_BRAKE; USER button
G	VSS	Ground
5V	-	5 V rail

Alternate functions are the multi-function options of the MG51FB9AE as given in the Nuvoton MG51 series datasheet. Functions shown as remap are selected in software. Every I/O pin can also be used as an interrupt source.

P0.5 carries the blue LED and P1.4 the USER button, so both pins have an on-board load; take this into account when using them for other functions.

6. Programming

The four-pin header on the top edge carries the Nuvoton ICP signals. Connect a Nu-Link or Nu-Link2 adapter and program with Nuvoton ICP Programming Tool, or use the same pins for on-chip debugging.

Header pin	Signal	MCU pin
RST	Reset	P2.0 / nRESET
CLK	ICP clock	P0.2 / ICE_CLK
DAT	ICP data	P1.6 / ICE_DAT
+5V	Target power	5 V rail

Toolchains: Keil C51, IAR Embedded Workbench for 8051, and SDCC with the Nuvoton 8051 BSP are all supported. Firmware can also be updated in the field over UART using the factory ISP loader in LDROM.

Note that P0.2 and P1.6 are shared between the ICP header and general I/O. Avoid loading these pins heavily if you want to keep in-circuit programming available.

7. Ordering information

Order code	Description
MN-MG51-DEV	MG51 Dev Board, assembled and tested, header strips not soldered
MN-MG51-DEV-H	MG51 Dev Board, assembled and tested, 2.54 mm header strips soldered

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This document describes revision 1.0 of the board. Specifications may change without notice. MG51 and NuMicro are trademarks of Nuvoton Technology Corporation; refer to the Nuvoton MG51 series datasheet for the authoritative device specification. The board is intended for evaluation and prototyping and is not qualified for use in life-support or safety-critical equipment.